Protective Coatings Hard and Soft Film Corrosion Preventive Compound – General Data

1. General

WARNING **THIS PROCEDURE USES MATERIALS THAT MAY BE DANGEROUS TO THE PERSONNEL'S HEALTH AND/OR THE ENVIRONMENT. SUFFICIENT PRECAUTIONS MUST BE TAKEN AT ANY STEP OF USE. IF NOT, THIS CAN CAUSE DAMAGE TO THE ENVIRONMENT AND DANGEROUS EFFECTS ON THE PERSONNEL'S HEALTH AND SAFETY.**

This standard specifies the procedures and requirements for the application of hard and soft film corrosion preventive compound to aircraft parts and assemblies.

A. Materials

- Hard Film Corrosion Preventive Compound to MIL–C–16173 Grade 1 (TECTYL 890, TECTYL 891, PETROTECT 1–X, NOX–RUST 501, ESGARD PL–1)
- Soft Film Corrosion Preventive Compound to MIL–C–16173 Grade 2 (TECTYL 502C, NOX–RUST 502, ESGARD PL–2, PETROTECT 2 OR LPS 3)
- Soft Film Corrosion Preventive Compound, Transparent, Water Displacing, Type I (colourless Boeshield T9 and LPS No. 3) and Type II (coloured Boeshield T9, ZC 023, ZCD–5023NS, COR–BAN35, LPS HARD COAT, PROTECTOR 100 and AV08)
- Solvent – DS108
- Solvent – NAPHTHA
- Solvent – Diestone DLS or HFP
- Solvent – Varsol
- Solvent – Toluol

B. Safety Precautions

1) Smoking, open flames or naked lights are prohibited in areas where spray application of corrosion preventive compound is being carried out.

2) The aircraft and aircraft major assemblies shall be electrically grounded during spraying operations.

3) Operators shall wear suitable protective respiratory equipment when applying corrosion preventive compound.

C. Preparation of Corrosion Preventive Compounds

WARNING **KEEP SOLVENT AWAY FROM FIRE AND OTHER SOURCES OF IGNITION, PROVIDE SUFFICIENT VENTILATION WHEN WORKING IN CONFINED SPACES, AND AVOID SKIN CONTACT WITH SOLVENT.**

1) Hard Film Compound

For brush, spray, dip, oil can or pour application, use as received.

Note Do not thin hard film, Grade 1 compounds, with solvents.

2) Soft Film Compound

For dip or brush application, dilute soft film compound with varsol to the consistency of thin syrup.

3) Soft Film Compound (Boeshield and LPS No. 3).

For brush or spray application, use as received from manufacturer.

2. Preparation of Parts

   A. Prepare Parts

      1) All operations, including welding, heat treatment, sand blasting, plating and painting, must be completed prior to applying corrosion preventive compound, except that open end tubes may be treated before painting, provided that sufficient time is allowed for the film to dry, and the outside of the tube is adequately cleaned prior to painting.

      2) Parts must be free from rust, oxides, etc., before applying corrosion preventive compound.

      3) Except for large parts or assemblies, degrease parts contaminated with shop oil or grease before applying corrosion preventative compound solvent clean large assemblies.

3. Application of Hard Film Compound (Grade 1)

   A. Open End Tubes

      1) Apply F13 Grade 1 hard film to the inside of open ended tubes as follows:

         a) Mask off any open holes or cut-outs. If the tube is open at both ends, mask off one end.

         b) Fill the tube with F13 Grade 1 hard film slowly to prevent any air bubble formation.

         c) Completely drain excess film from the inside of the tube.

         d) Position the part vertically and allow to dry for at least 4 hours.

         e) Ensure that the inside of the tube is completely covered with the film (i.e., the film should be continuous and uniformly black). Re-coat if necessary.

   B. Sealed Tubes

      1) Sealed or closed tubes or parts must be treated by pressure oiling through the drilled holes provided for this purpose. When sufficient compound is inside, seal the holes temporarily with tape. Rotate and upend the part to ensure that the film contacts all interior surfaces.

      2) Remove tape and allow excess compound to drain completely. Multi-sectioned parts must be turned frequently to ensure drainage from all low points and pockets.

      3) When the parts are thoroughly drained, seal the drill holes with self-plugging cherry rivets.

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      4) Clean all excess compound from exterior surfaces by wiping with a clean cloth dampened with NAPHTHA. Wipe dry with a clean cloth before solvent evaporates.

   C. Small Parts

      1) Small parts and assemblies on which the surfaces to be treated are readily accessible may be treated by dipping or brush coating with a thin uniform film of compound.

4. Application of Soft Film Compound (Grade 2)

   A. Small Parts

      Small parts and assemblies may be treated by dipping or brush coating with a thin uniform film of compound.

5. Application of Soft Film Compound (Grade 3 Type I and Type II)

   A. Small Parts

      Small parts and assemblies may be treated by brush or spray coating using mist action hand trigger spray bottles or standard paint spraying equipment to achieve a thin uniform film of compound.

   B. Large Parts

      Large parts or areas shall be treated by spraying using mist action hand trigger spray bottles, standard paint spraying equipment or extension spray equipment to achieve a thin film of compound.

      Note When spraying compound on to the interior of an aircraft, the following items shall be suitably masked or shielded from direct spray or over spray contamination:

      - Black boxes with cooling holes
      - Windows
      - Oxygen tubes and fittings
      - All other system tubing open ends
      - Control cables and associated hardware
      - Systems components

      Note A light over spray on the following items is not detrimental, but direct spray shall be avoided:

      - Bearings
      - Pulleys
      - Rubber items
      - Non-oxygen system tubes

      Note Subsequent to spray application, enclosed areas shall be allowed to dry in accordance with one of the following schedules before resuming work on the aircraft or assembly:

      - Drying Time:

        • 1 hour with forced ventilation
        • 6-8 hours with natural ventilation

6. Removal of Film

   Hard and soft film may be removed by washing the part with DS108, Diestone DLS or Diestone HFP.

7. Touch-Up and Repair

   A. If corrosion preventive compound has been removed from treated surfaces, touch-up these areas with the same type of compound as originally used on the parts. It is acceptable to use aerosol cans, if available, containing the same compound for touch-up.

   B. If parts have been reworked after application of corrosion preventive compound, ensure that the surfaces are free of swarf or chips before re-application of compound.

8. Requirements

   A. All areas must be completely covered with a continuous film of compound.

   B. Insides of open tubes must be examined for continuity of coating by examining the bore against a light. Uncovered areas will show up as dark patches.

C. The film thickness shall be such that the applied coating is continuous, but running of the compound is kept to a minimum.